Michael Yuxiang Tang

Email: myt001@ucsd.edu

Education

University of California, San Diego Sept. 2023 - present
Ph.D., Electrical Engineering (Intelligent Systems, Robotics, Control)
Advisor: Jorge Poveda
M.S. degree awarded in June 2025

University of California, San Diego Sept. 2019 - June 2023
B.S., Electrical Engineering
B.S., Applied Mathematics

Professional Experience

Graduate Student Researcher, Air Force Research Laboratory May 2026 - Aug. 2026

Selected Awards and Honors

Best Student Paper Award, American Control Conference, 2026
NSF Graduate Research Fellowship, 2025
Best Student Paper Award Finalist, American Control Conference, 2024
UCSD ECE Department Fellowship, 2023
Cum Laude Honors (3.928 GPA) from UCSD, 2023

Journal Publications

[J4] **M. Tang**, M. Krstic, J. I. Poveda, "A Lyapunov-Based Small-Gain Theorem for Fixed-Time ISS: Theory, Optimization, and Games", IEEE Transactions on Automatic Control, February 2027

[J3] **M. Tang**, M. Abdelgalil, J. I. Poveda, "A Separation Principle for Stochastic Feedback Optimization", IEEE Control Systems Letters, June 2026

[J2] **M. Tang**, M. Krstic, J. I. Poveda, "Fixed-Time Input-to-State Stability for Singularly Perturbed Systems via Composite Lyapunov Functions", IEEE Transactions on Automatic Control, April 2026

[J1] **M. Tang**, U. Javed, X. Chen, M. Krstic and J. I. Poveda, "Deception in Nash Equilibrium Seeking", IEEE Transactions on Automatic Control, December 2025

Conference Proceedings

[C9] **M. Tang**, M. Abdelgalil, J. I. Poveda, "The Deceptive Bandit Problem: Exploratory Coupling and the Fragility of Multi-Agent Learning", IEEE Conference on Decision and Control, 2026

[C8] K. Vamvoudakis, J. I. Poveda, S. Anderson, J. P. Hespanha, M. Krstic, I. R. Petersen, **M. Tang**, A. Wadi, Q. Zhu, "A Tutorial for Foundations in AI and Quantum Systems: From Control and Game-Theory to Stochastic Deception", American Control Conference, 2026

[C7] **M. Tang**, M. Krstic, J. I. Poveda, "A Lyapunov-Based Small-Gain Theorem for Fixed-Time Stability", American Control Conference, 2026
Best Student Paper Award

[C6] **M. Tang**, M. Krstic, J. I. Poveda, "On Sufficient Lyapunov Conditions for Fixed-Time Stability of Hybrid Dynamical Systems", IEEE Conference on Decision and Control, 2025

[C5] **M. Tang**, M. Krstic, J. I. Poveda, "Stochastic Real-Time Deception in Nash Equilibrium Seeking for Games with Quadratic Payoffs", 7th Learning for Dynamics and Control Conference, 2025

[C4] K. Vamvoudakis, F. Fotiadis, T. Başar, V. Gupta, J. I. Poveda, **M. Tang**, M. Krstic, Q. Zhu, "Deception in Game Theory and Control: A Tutorial", American Control Conference, 2025.

[C3] **M. Tang**, M. Krstic, J. I. Poveda, "Deception in Oligopoly Games via Adaptive Nash Seeking Systems", 13th International Conference on Game Theory for Networks, 2025

[C2] **M. Tang**, M. Krstic, J. I. Poveda, "Filterless Fixed-Time Extremum Seeking for Scalar Quadratic Maps", IEEE Conference on Decision and Control, 2024

[C1] **M. Tang**, M. Krstic, J. I. Poveda, "On Fixed-Time Stability for a Class of Singularly Perturbed Systems using Composite Lyapunov functions", American Control Conference, 2024
Best Student Paper Award Finalist

Preprints/Submitted

[S1] **M. Tang**,

Travel Support Awards

Student Travel Grant, American Control Conference, 2026
Student Travel and Workshop Support Award, IEEE Conference on Decision and Control, 2025
Student Travel and Workshop Support Award, IEEE Conference on Decision and Control, 2024
Student Travel Grant, American Control Conference, 2024

Invited Talks

05/2025: "Fixed-Time ISS in Singular Perturbations", UNAM Variable Structure Systems Seminar

03/2024: "Deceptive Nash Equilibrium Seeking", UCSD Control and Optimization Seminar

Conference and Workshop Presentations

05/2026: "A Lyapunov-Based Small-Gain Theorem for Fixed-Time Stability", American Control Conference, New Orleans, Louisiana

05/2026: "Stochastic Deception in Noncooperative Games", Tutorial Session (Co-chair), American Control Conference, New Orleans, Louisiana

12/2025: "On Sufficient Lyapunov Conditions for Fixed-Time Stability of Hybrid Dynamical Systems", IEEE Conference on Decision and Control, Rio de Janeiro, Brazil

06/2025: "Stochastic Real-Time Deception in Nash Equilibrium Seeking for Games with Quadratic Payoffs" (Poster), 7th Learning for Dynamics and Control Conference, Ann Arbor, Michigan

04/2025: "Recent Advances in the Study of Fixed-time Stability", 45th Southern California Control Workshop, UCSD

03/2025: "Deception in Oligopoly Games via Adaptive Nash Seeking Systems", 13th International Conference on Game Theory for Networks, Cambridge, UK

12/2024: "Filterless Fixed-Time Extremum Seeking for Scalar Quadratic Maps", IEEE Conference on Decision and Control, Milan, Italy

07/2024: "On Fixed-Time Stability for a Class of Singularly Perturbed Systems using Composite Lyapunov functions", American Control Conference, Denver, Colorado

08/2021: "OCT-Angiography Processing for Disease Detection and Classification", UCSD Summer Research Conference

Teaching

Spring 2026: Co-lecturer for ECE 272A (Graduate Linear Systems), UCSD

Winter 2026: Tutor for ECE 285 (Stochastic Hybrid Dynamical Systems), UCSD

Academic Service

Conference Reviewer: American Control Conference, IEEE Conference on Decision and Control, IFAC Conference on Networked Systems

Journal Reviewer: IEEE Transactions on Automatic Control; IEEE Transactions on Control of Network Systems; IEEE Control Systems Letters; Automatica; International Journal on Nonlinear and Robust Control; Nonlinear Analysis Hybrid Systems